

Design of a Compact, Ultra-Broadband 3-Section Impedance Matching Network

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Abstract—This report details the design and optimization of a multi-section microstrip impedance matching network. The objective is to match a $200\ \Omega$ lumped resistor load to a $50\ \Omega$ source at a center frequency of 2 GHz. To maximize bandwidth while satisfying strict fabrication constraints, a modified 3-section Chebyshev transformer was synthesized. Through iterative schematic tuning and layout optimization, the final Momentum electromagnetic (EM) simulation achieved a 93.6% fractional bandwidth with a compact footprint of $25.8 \times 27.1\ \text{mm}^2$, exceeding all design specifications.

Index Terms—Chebyshev matching, impedance matching, meandering, impedance transformer

I. INTRODUCTION

IMPEDANCE matching is fundamental in RF engineering to ensure maximum power transfer. This project requires matching a $200\ \Omega$ 0805 SMD resistor to a $50\ \Omega$ source at 2 GHz using a Rogers RT/duroid 5880 substrate ($\epsilon_r = 2.2$, $H = 1.57\ \text{mm}$, $\tan \delta = 0.0009$). The design objectives require either a minimum 40% fractional bandwidth with $\geq 20\ \text{dB}$ return loss, or a compact size of $\leq 28 \times 28\ \text{mm}^2$.

II. RATIONALE BEHIND DESIGN CHOICES

A. Topology Selection

To achieve a broadband response well beyond the 40% requirement, a multi-section transformer was chosen over narrow-band stub matching. A Chebyshev polynomial response was targeted for its optimal equal-ripple passband behavior. Although a 2-section ($N = 2$) design satisfies the minimum criteria, an $N = 3$ architecture was pursued to explore the limits of the maximum achievable bandwidth.

B. Theoretical Calculations and Fabrication Constraints

According to standard Chebyshev tables for $\Gamma_m = 0.05$ and an impedance ratio of 4, the normalized impedances for a 3-section transformer are $Z_1/Z_0 = 1.2662$, $Z_2/Z_0 = 2.0000$, and $Z_3/Z_0 = 3.1591$. For a $50\ \Omega$ system, the impedance closest to the load must be $Z_3 = 157.95\ \Omega$. However, ADS LineCalc indicates that a $157.95\ \Omega$ microstrip line on the specified substrate has a width of only $0.37\ \text{mm}$, which violates the strict fabrication rule of *Minimum T-line width* $\geq 0.5\ \text{mm}$.

To resolve this, the maximum allowable impedance satisfying $W \geq 0.5\ \text{mm}$ was calculated via LineCalc to be $145\ \Omega$. To maintain a symmetric reflection profile, the center section

impedance was set to the geometric mean of the source and the load:

$$Z_2 = \sqrt{Z_0 Z_L} = \sqrt{50 \times 200} = 100\ \Omega \quad (1)$$

According to the theory of small reflections for symmetric networks, the local reflection coefficients must be symmetric ($\gamma_0 = \gamma_3$ and $\gamma_1 = \gamma_2$). The coefficient γ_2 is calculated as:

$$\gamma_2 = \frac{Z_3 - Z_2}{Z_3 + Z_2} = \frac{145 - 100}{145 + 100} \approx 0.1837 \quad (2)$$

Setting $\gamma_1 = \gamma_2 \approx 0.1837$, the required impedance for the first section Z_1 is derived:

$$\gamma_1 = \frac{Z_2 - Z_1}{Z_2 + Z_1} \implies Z_1 = Z_2 \frac{1 - \gamma_1}{1 + \gamma_1} \approx 68.97\ \Omega \quad (3)$$

Based on these modified characteristic impedances, the theoretical initial dimensions synthesized using LineCalc at 2 GHz are:

- $Z_1 = 68.97\ \Omega$: $W_1 = 2.87\ \text{mm}$, $L_1 = 27.72\ \text{mm}$
- $Z_2 = 100\ \Omega$: $W_2 = 1.38\ \text{mm}$, $L_2 = 28.25\ \text{mm}$
- $Z_3 = 145\ \Omega$: $W_3 = 0.504\ \text{mm}$, $L_3 = 28.73\ \text{mm}$

III. SCHEMATIC DESIGN AND TUNING

A. Straight Transformer Design and Tuning

The calculated dimensions were initially implemented in an ADS schematic using straight microstrip lines (MLIN). Due to the modifications made to the design parameters, initial simulations showed the S_{11} curve slightly deviating from the $-20\ \text{dB}$ limit at the ripple peaks. The ADS Tuning tool was utilized to fine-tune the size of each section, lowering the return loss and re-centering the response to achieve optimal matching at 2 GHz (Fig. 1).

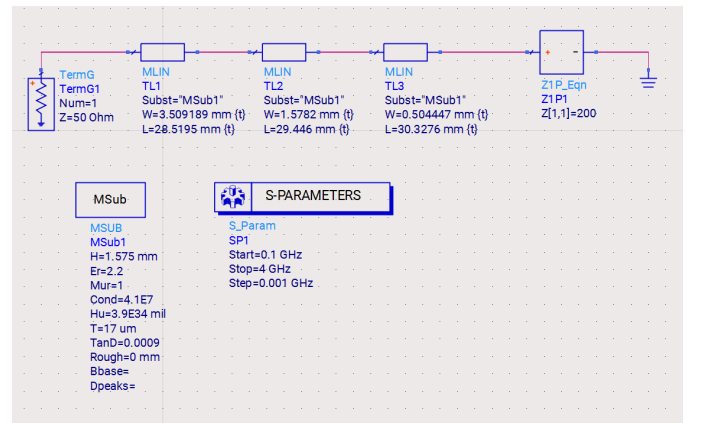


Fig. 1. Schematic of the 3-section transformer after initial tuning.

B. Curved Transformer Design and Tuning

A linear connection of the three sections results in a total length exceeding 84 mm. To satisfy the $28 \times 28 \text{ mm}^2$ constraint, the network was meandered using MCURVE elements. The total electrical length of each stage was conserved by subtracting the physical arc length of the curves from the adjacent straight MLIN segments. The meandered structure was carefully routed to maintain sufficient spacing between parallel traces to minimize unwanted coupling. Subsequent tuning was performed to absorb the parasitic reactances introduced by the bends (Fig. 2).

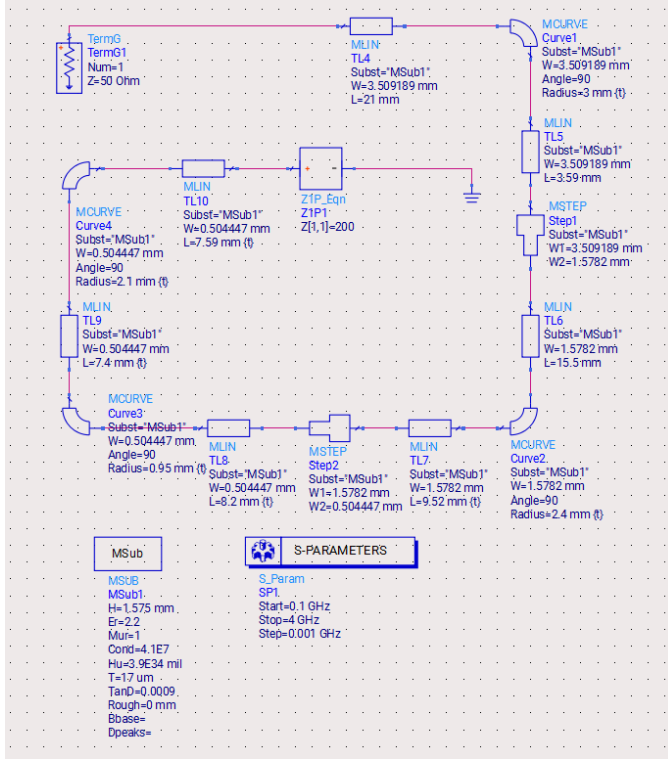


Fig. 2. Schematic of the meandered 3-section transformer after tuning.

IV. LAYOUT IMPLEMENTATION

For the physical layout, mounting pads for the 0805 resistor were designed with dimensions $W = 0.7 \text{ mm}$ and $L = 1.3 \text{ mm}$, separated by a 1.7 mm gap. A 0.4 mm diameter via was added to ground the load. The substrate stackup configured for the Momentum simulation, explicitly defining the Rogers 5880 dielectric layer and the via hole, is depicted in Fig. 3.

An iterative co-simulation approach was employed: initial Momentum EM simulations revealed frequency shifts and performance degradation. The primary factors contributing to this deviation were identified as the unwanted electromagnetic coupling between the closely meandered transmission line segments and the parasitics from the physical pads and grounding via. To mitigate these effects, the schematic parameters were iteratively adjusted to pre-compensate for the parasitic reactances and maximize the spatial separation between adjacent traces within the constrained footprint. The layout

was subsequently updated and re-simulated. This iterative optimization cycle was repeated until the EM-simulated S_{11} fully complied with the -20 dB specification across the entire wideband region (Fig. 4). A 3D perspective of the simulated electromagnetic structure, highlighting the port assignments and the grounding via, is presented in Fig. 5.

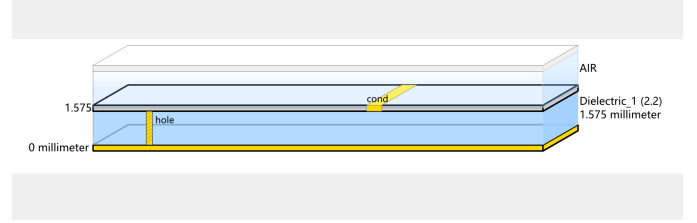


Fig. 3. Momentum substrate stackup defining the dielectric layer and grounding via.

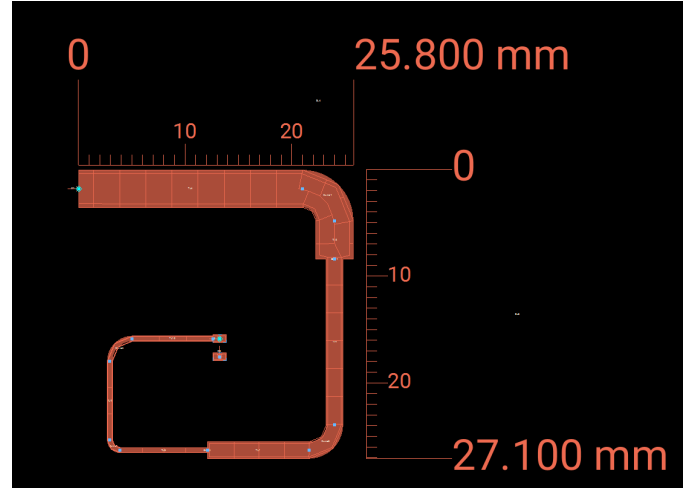


Fig. 4. Final meandered layout of the matching network ($25.8 \times 27.1 \text{ mm}^2$).

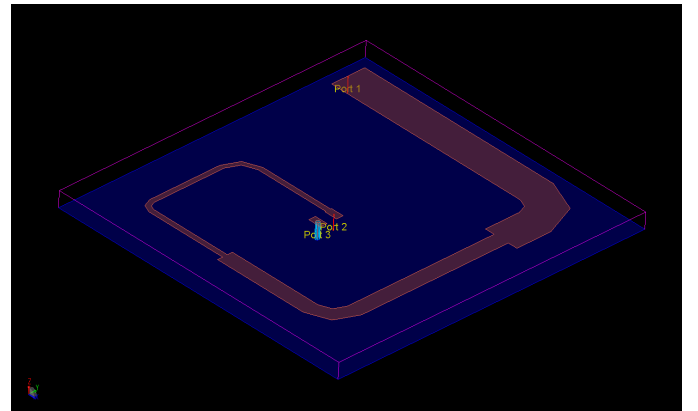


Fig. 5. 3D view of the matching network layout in Momentum.

V. SIMULATION RESULTS AND DISCUSSION

A. EM vs. Schematic Comparison

Fig. 6 presents the overlaid S_{11} results of the final schematic (blue trace) and Momentum (red trace) simulations. The two

curves exhibit excellent agreement with slight variations in the depth and position of the nulls. The Momentum simulation shows that the design maintains a highly matched return loss of -25.5 dB at the 2 GHz center frequency. The 20 dB bandwidth covers 1.058 GHz to 2.931 GHz, yielding an outstanding fractional bandwidth of approximately 93.6%.

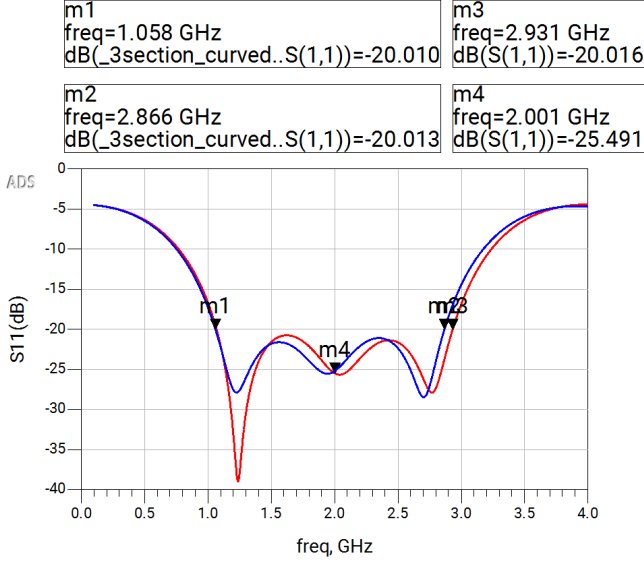


Fig. 6. Comparison of S_{11} between schematic and Momentum EM simulation.

B. Calculation of Expected Loss

The expected power loss of the passive matching network is evaluated using the principle of energy conservation, defined as:

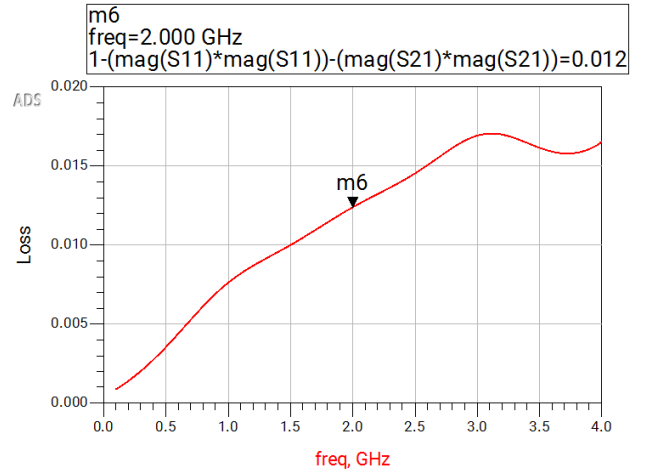
$$\text{Loss} = 1 - |S_{11}|^2 - |S_{21}|^2 \quad (4)$$

For the schematic simulation (Fig. 7a), the evaluated power dissipation at the 2 GHz center frequency is approximately 0.012, translating to a power loss of 1.2%.

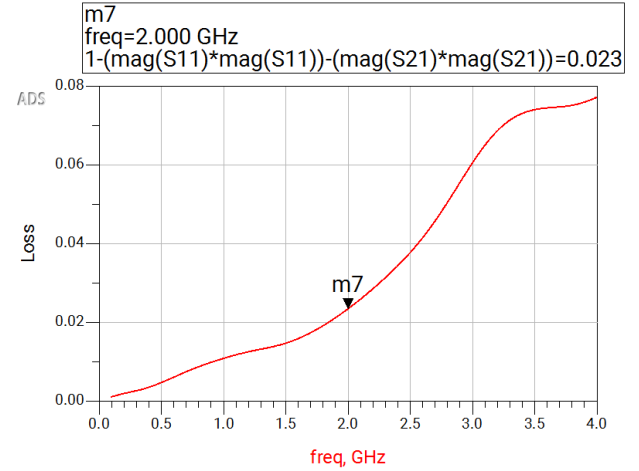
For the Momentum simulation (Fig. 7b), the loss at 2 GHz is approximately 0.023 (or 2.3%). The slightly higher loss in the Momentum simulation accurately reflects the additional ohmic losses, skin effects, and dielectric dissipation introduced by the physical copper traces and the Rogers 5880 substrate, which are not fully captured by ideal schematic models. Nevertheless, an overall transmission efficiency of 97.7% confirms the excellent low-loss characteristics of the final layout.

VI. CONCLUSION

A highly compact, 3-section Chebyshev matching network was successfully designed and validated. By mathematically reconstructing the impedance profile to respect the 0.5 mm fabrication limit and utilizing meandering techniques for miniaturization, the final Momentum layout achieved an exceptional 93.6% fractional bandwidth within a 25.8×27.1 mm² footprint. All performance metrics met or exceeded the project specifications.



(a) Schematic Simulation



(b) Momentum EM Simulation

Fig. 7. Simulated total power loss ratio evaluated directly using S-parameter magnitudes.

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